

# Comparison of Numerical Methods with Fourier Series Solution

## **Contents**

# 1 Introduction

This report presents a visual comparison between various numerical methods and the Fourier series solution for an RL circuit with a PWM input. The numerical methods compared are:

- Euler Method
- RK2 Method (2nd order Runge-Kutta)
- RK4 Method (4th order Runge-Kutta)
- Reverse Euler Method
- Trapezoidal Method

## 2 Parameter Configurations

The comparisons are made across different parameter configurations:

- Resistance (R): 1000  $\Omega$ , 100  $\Omega$ , 1  $\Omega$ , 0.001  $\Omega$
- Inductance (L): 1 H
- Time step (h): T/10, T/1000 (where T = 0.1s is the time period)
- Duty cycle ( $\alpha$ ): 0.4
- Amplitude (A): 10.0

## 3 Results for R = 1000 $\Omega$

### 3.1 Time Step h = T/10

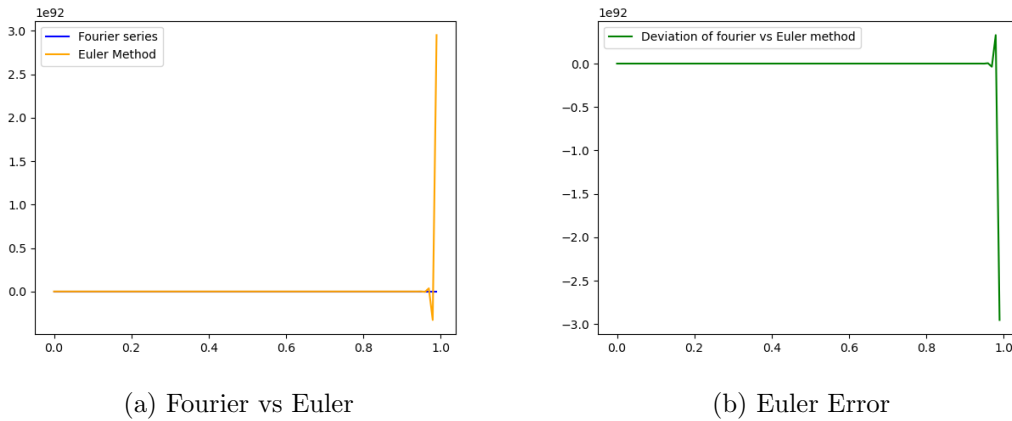
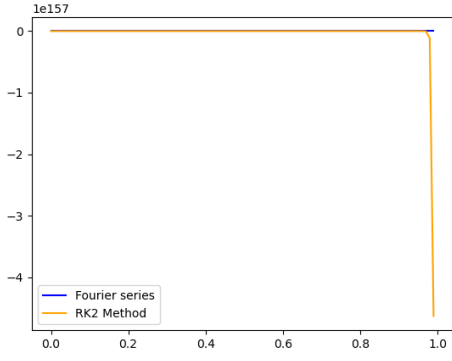
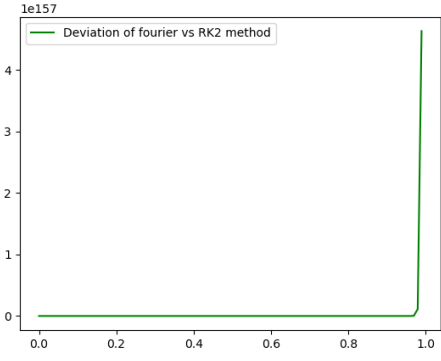


Figure 1: Euler Method Comparison (R = 1000  $\Omega$ , L = 1 H, h = T/10)

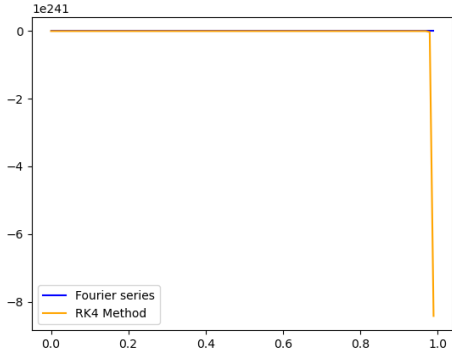


(a) Fourier vs RK2

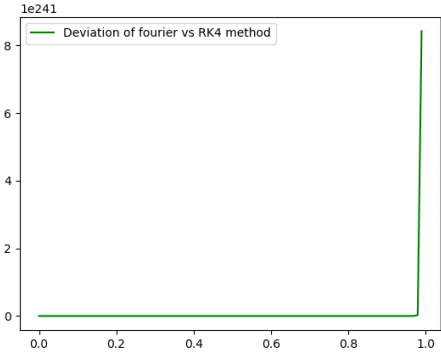


(b) RK2 Error

Figure 2: RK2 Method Comparison ( $R = 1000 \Omega$ ,  $L = 1 \text{ H}$ ,  $h = T/10$ )

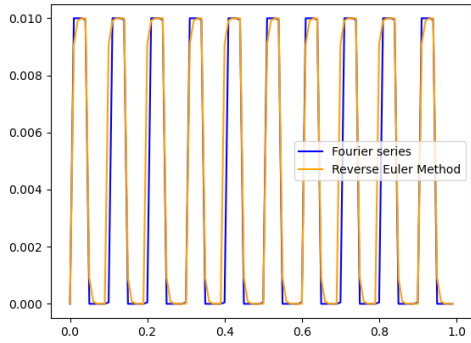


(a) Fourier vs RK4

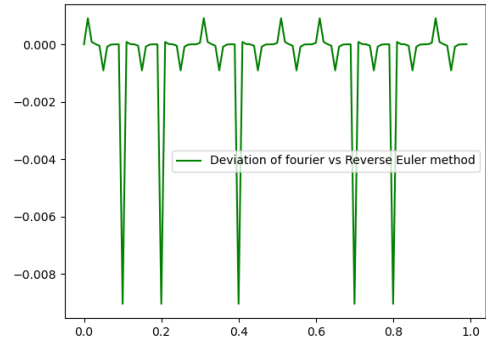


(b) RK4 Error

Figure 3: RK4 Method Comparison ( $R = 1000 \Omega$ ,  $L = 1 \text{ H}$ ,  $h = T/10$ )

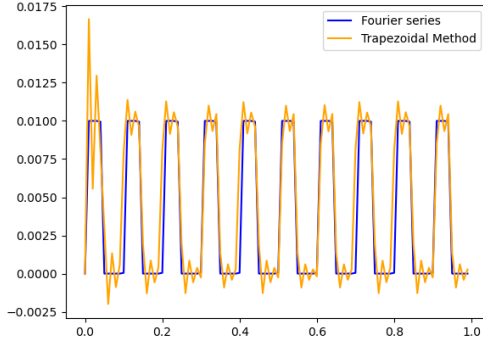


(a) Fourier vs Reverse Euler

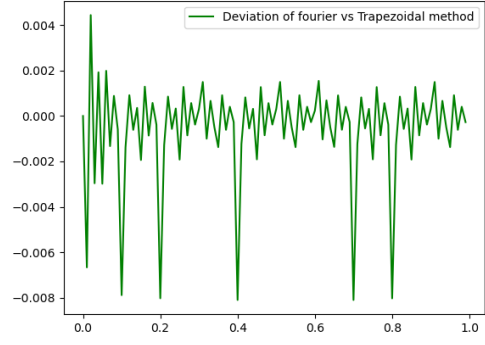


(b) Reverse Euler Error

Figure 4: Reverse Euler Method Comparison ( $R = 1000 \Omega$ ,  $L = 1 \text{ H}$ ,  $h = T/10$ )

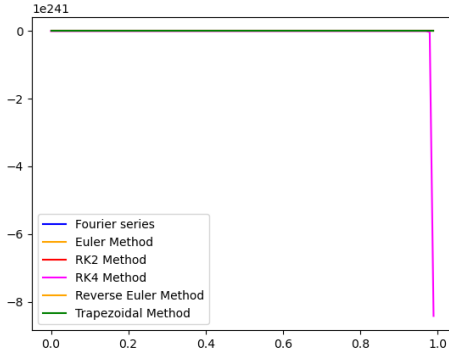


(a) Fourier vs Trapezoidal

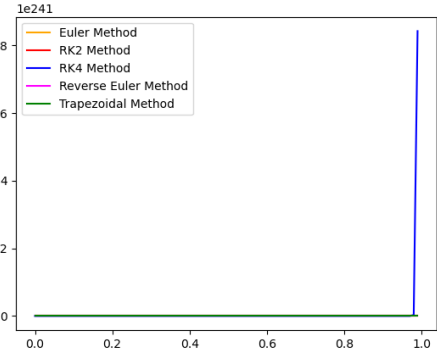


(b) Trapezoidal Error

Figure 5: Trapezoidal Method Comparison ( $R = 1000 \Omega$ ,  $L = 1 \text{ H}$ ,  $h = T/10$ )



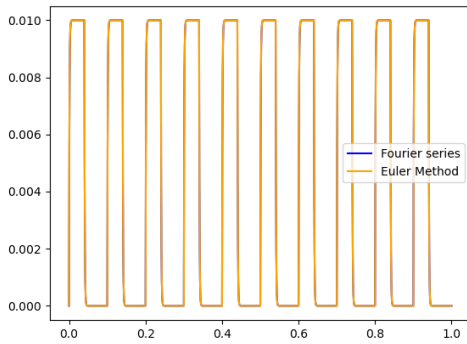
(a) All Methods Comparison



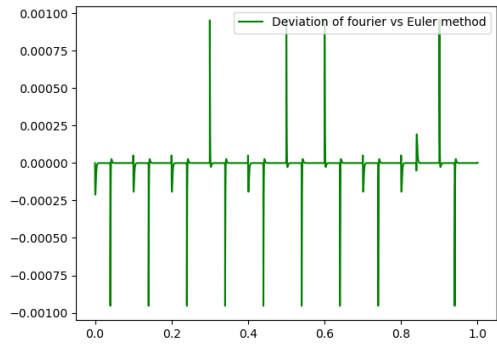
(b) All Methods Error

Figure 6: Comparison of All Methods ( $R = 1000 \Omega$ ,  $L = 1 \text{ H}$ ,  $h = T/10$ )

### 3.2 Time Step $h = T/1000$

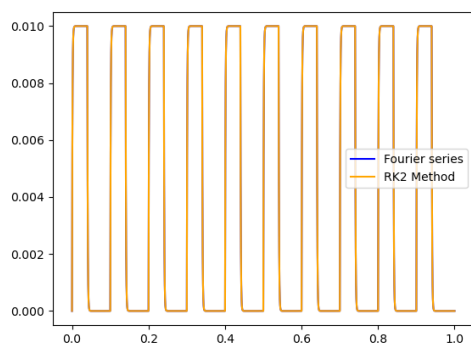


(a) Fourier vs Euler

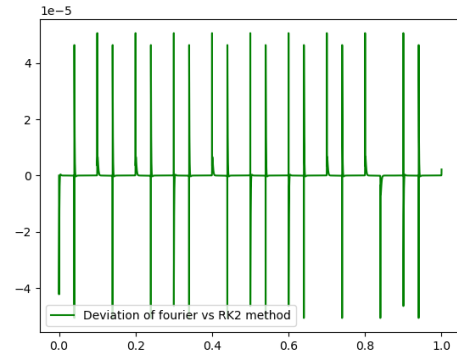


(b) Euler Error

Figure 7: Euler Method Comparison ( $R = 1000 \Omega$ ,  $L = 1 \text{ H}$ ,  $h = T/1000$ )

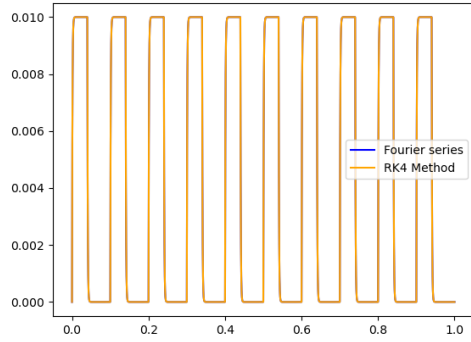


(a) Fourier vs RK2

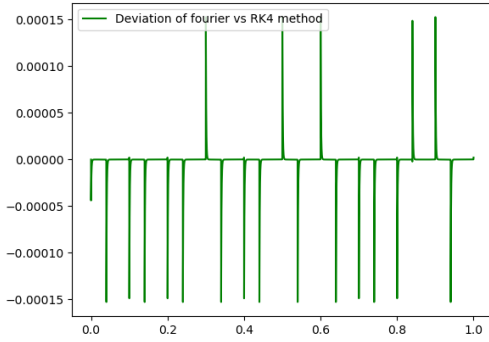


(b) RK2 Error

Figure 8: RK2 Method Comparison ( $R = 1000 \Omega$ ,  $L = 1 \text{ H}$ ,  $h = T/1000$ )

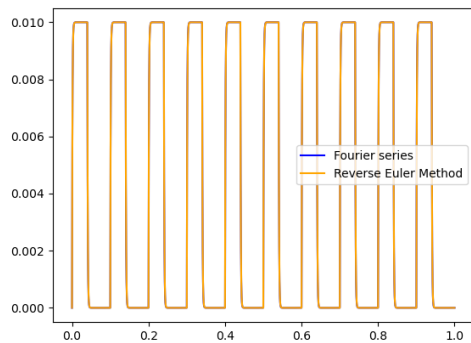


(a) Fourier vs RK4

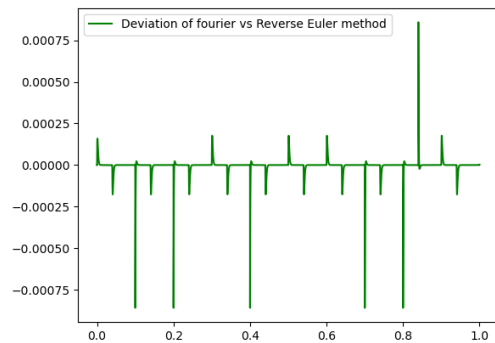


(b) RK4 Error

Figure 9: RK4 Method Comparison ( $R = 1000 \Omega$ ,  $L = 1 \text{ H}$ ,  $h = T/1000$ )

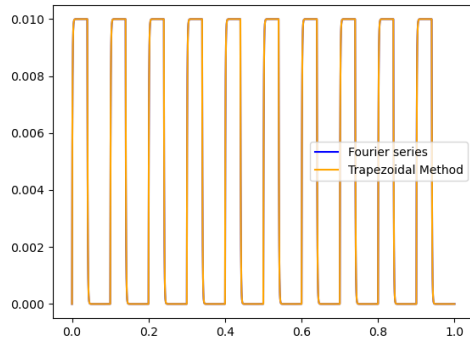


(a) Fourier vs Reverse Euler

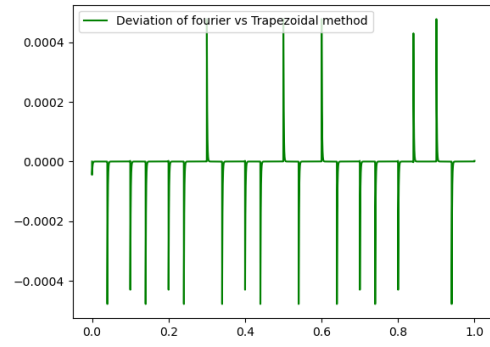


(b) Reverse Euler Error

Figure 10: Reverse Euler Method Comparison ( $R = 1000 \Omega$ ,  $L = 1 \text{ H}$ ,  $h = T/1000$ )

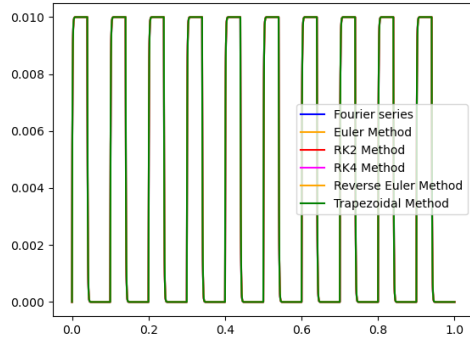


(a) Fourier vs Trapezoidal

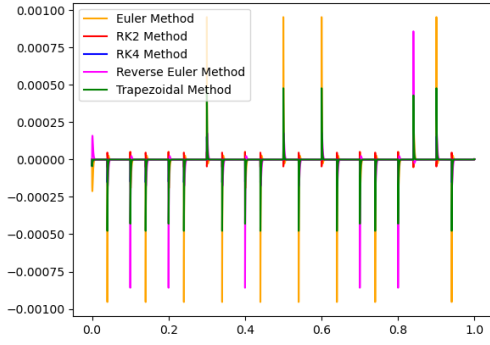


(b) Trapezoidal Error

Figure 11: Trapezoidal Method Comparison ( $R = 1000 \Omega$ ,  $L = 1 \text{ H}$ ,  $h = T/1000$ )



(a) All Methods Comparison



(b) All Methods Error

Figure 12: Comparison of All Methods ( $R = 1000 \Omega$ ,  $L = 1 \text{ H}$ ,  $h = T/1000$ )

## 4 Results for $R = 100 \Omega$

### 4.1 Time Step $h = T/10$

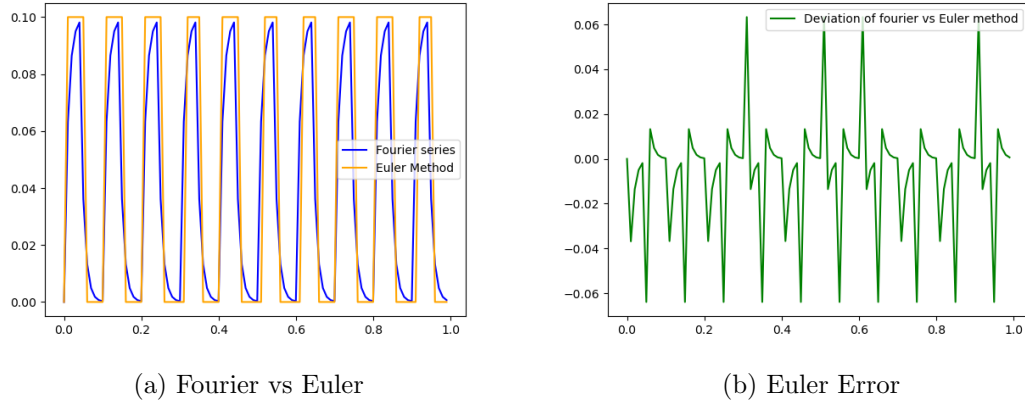


Figure 13: Euler Method Comparison ( $R = 100 \Omega$ ,  $L = 1 \text{ H}$ ,  $h = T/10$ )

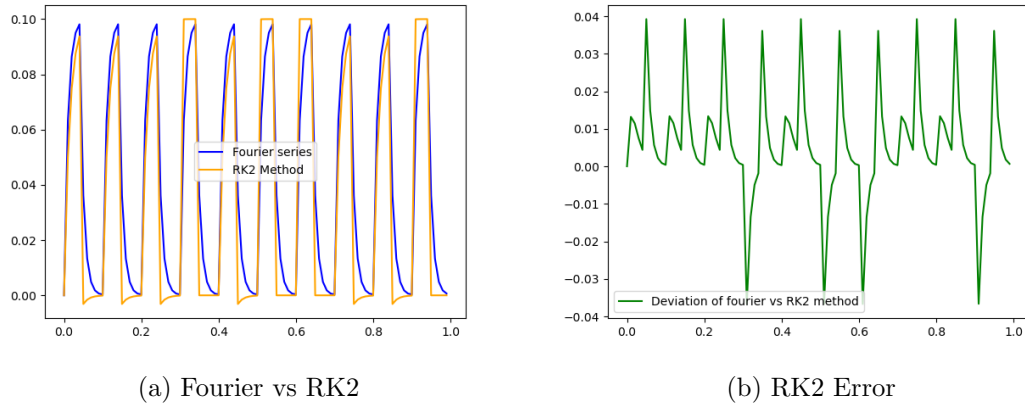
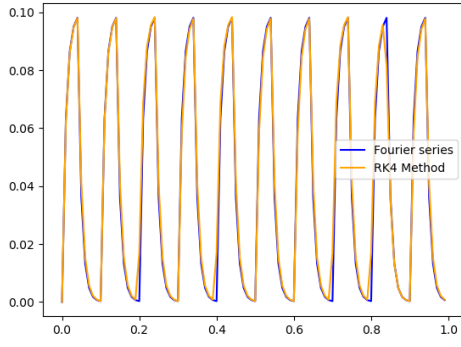
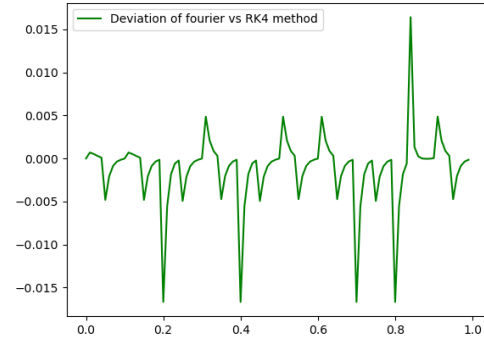


Figure 14: RK2 Method Comparison ( $R = 100 \Omega$ ,  $L = 1 \text{ H}$ ,  $h = T/10$ )

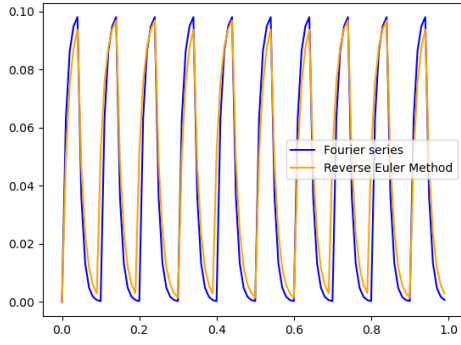


(a) Fourier vs RK4

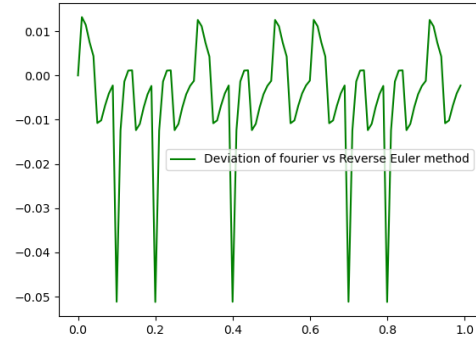


(b) RK4 Error

Figure 15: RK4 Method Comparison ( $R = 100 \Omega$ ,  $L = 1 \text{ H}$ ,  $h = T/10$ )

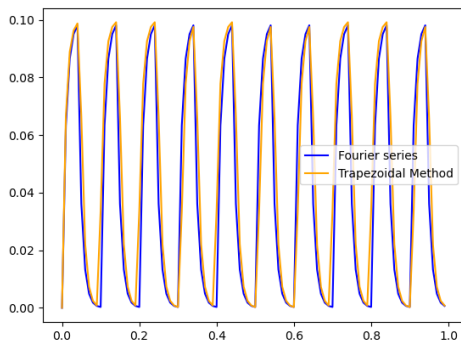


(a) Fourier vs Reverse Euler

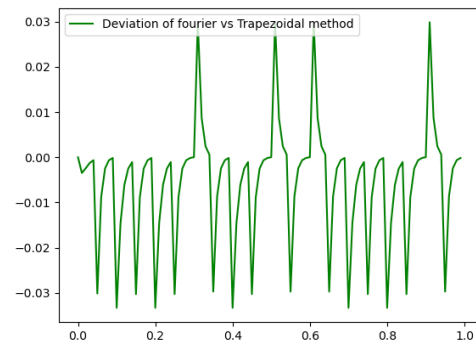


(b) Reverse Euler Error

Figure 16: Reverse Euler Method Comparison ( $R = 100 \Omega$ ,  $L = 1 \text{ H}$ ,  $h = T/10$ )

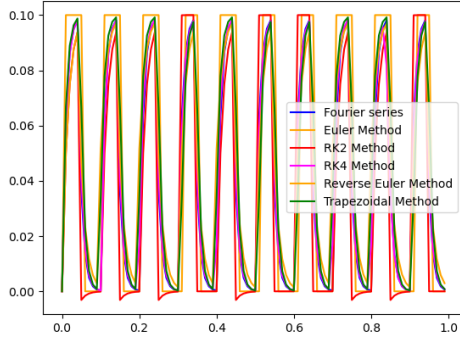


(a) Fourier vs Trapezoidal

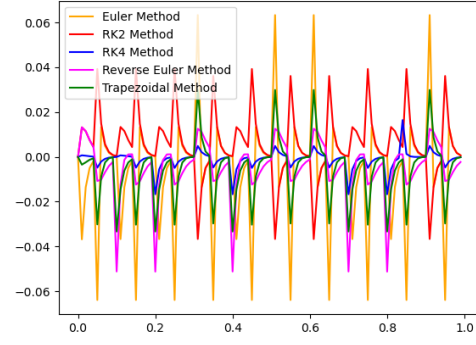


(b) Trapezoidal Error

Figure 17: Trapezoidal Method Comparison ( $R = 100 \Omega$ ,  $L = 1 \text{ H}$ ,  $h = T/10$ )



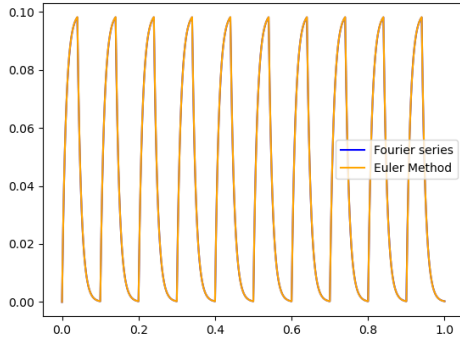
(a) All Methods Comparison



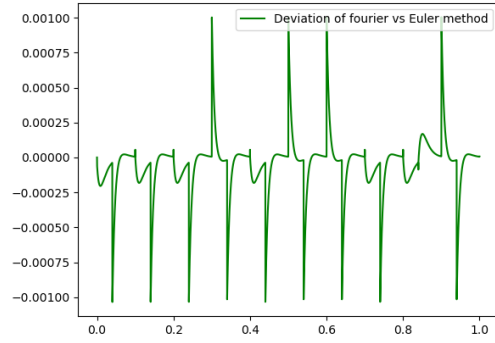
(b) All Methods Error

Figure 18: Comparison of All Methods ( $R = 100 \Omega$ ,  $L = 1 \text{ H}$ ,  $h = T/10$ )

## 4.2 Time Step $h = T/1000$

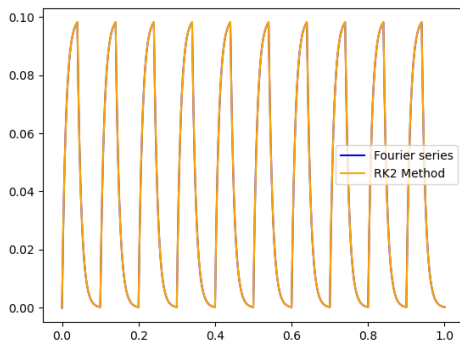


(a) Fourier vs Euler

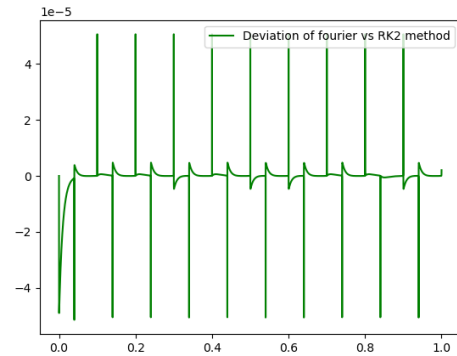


(b) Euler Error

Figure 19: Euler Method Comparison ( $R = 100 \Omega$ ,  $L = 1 \text{ H}$ ,  $h = T/1000$ )

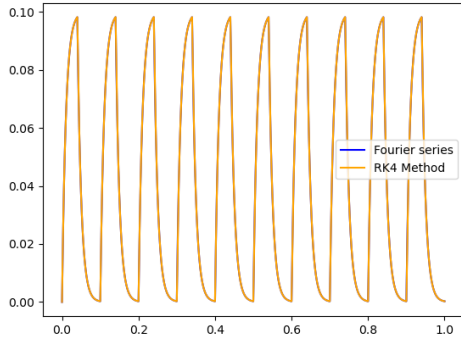


(a) Fourier vs RK2

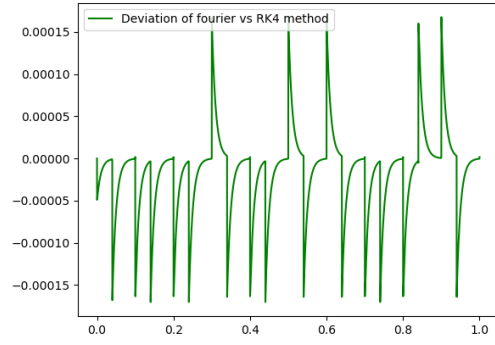


(b) RK2 Error

Figure 20: RK2 Method Comparison ( $R = 100 \Omega$ ,  $L = 1 \text{ H}$ ,  $h = T/1000$ )

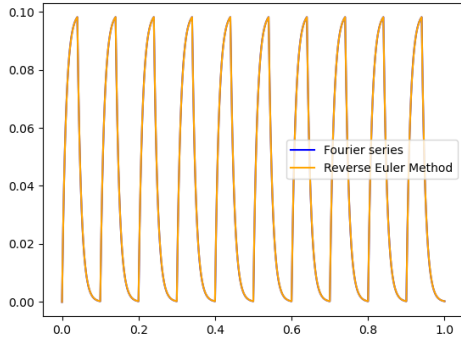


(a) Fourier vs RK4

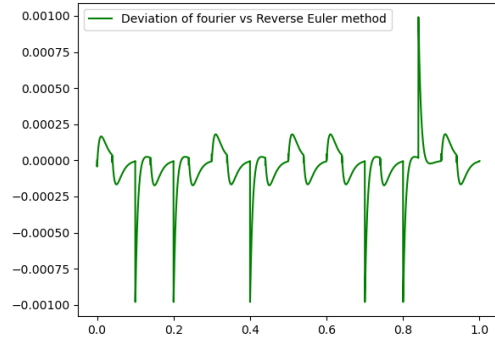


(b) RK4 Error

Figure 21: RK4 Method Comparison ( $R = 100 \Omega$ ,  $L = 1 \text{ H}$ ,  $h = T/1000$ )

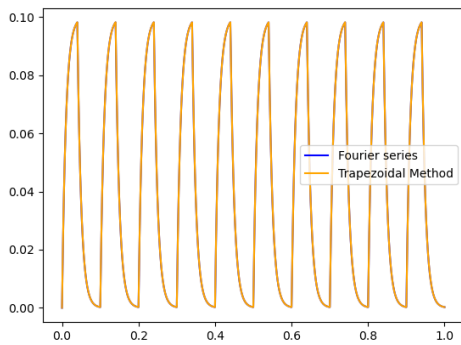


(a) Fourier vs Reverse Euler

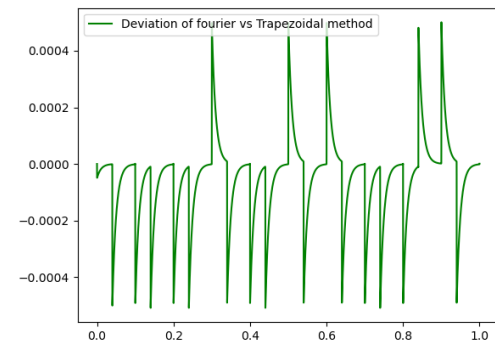


(b) Reverse Euler Error

Figure 22: Reverse Euler Method Comparison ( $R = 100 \Omega$ ,  $L = 1 \text{ H}$ ,  $h = T/1000$ )

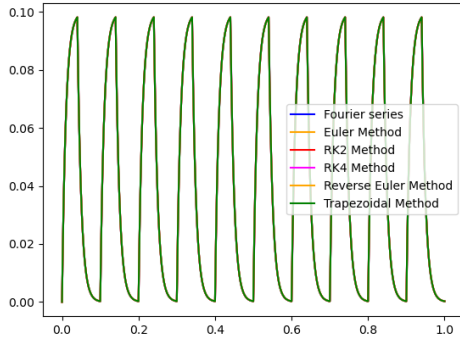


(a) Fourier vs Trapezoidal

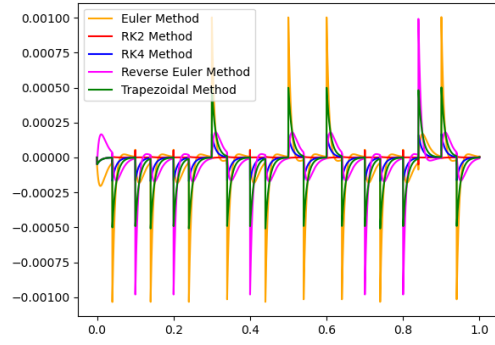


(b) Trapezoidal Error

Figure 23: Trapezoidal Method Comparison ( $R = 100 \Omega$ ,  $L = 1 \text{ H}$ ,  $h = T/1000$ )



(a) All Methods Comparison

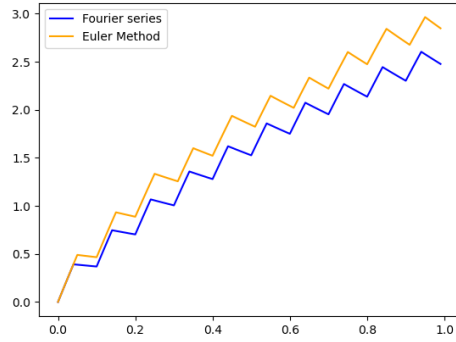


(b) All Methods Error

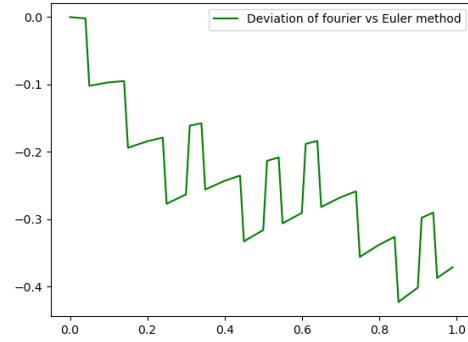
Figure 24: Comparison of All Methods ( $R = 100 \Omega$ ,  $L = 1 \text{ H}$ ,  $h = T/1000$ )

## 5 Results for $R = 1 \Omega$

### 5.1 Time Step $h = T/10$

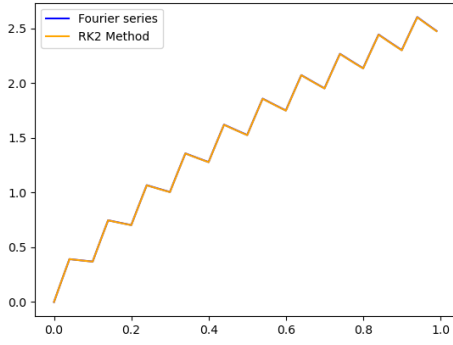


(a) Fourier vs Euler

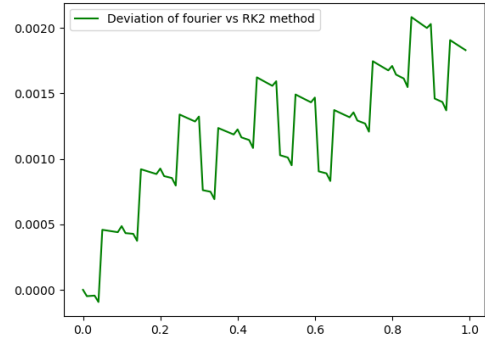


(b) Euler Error

Figure 25: Euler Method Comparison ( $R = 1 \Omega$ ,  $L = 1 \text{ H}$ ,  $h = T/10$ )

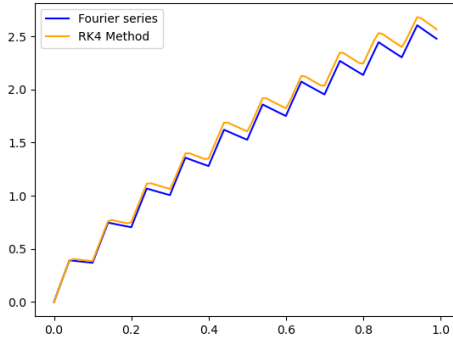


(a) Fourier vs RK2

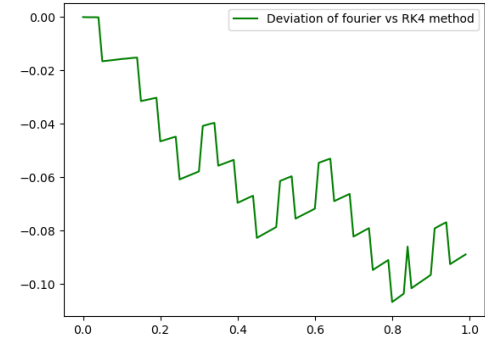


(b) RK2 Error

Figure 26: RK2 Method Comparison ( $R = 1 \Omega$ ,  $L = 1 H$ ,  $h = T/10$ )

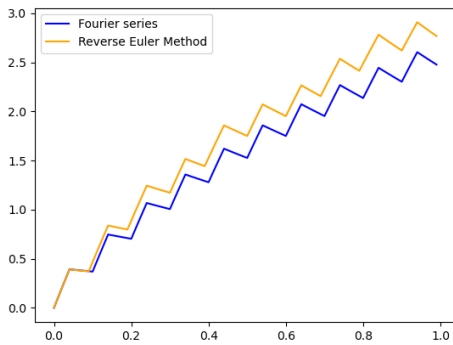


(a) Fourier vs RK4

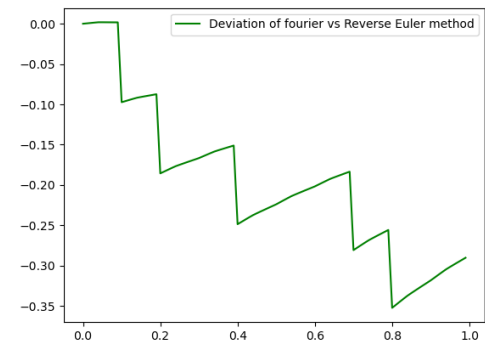


(b) RK4 Error

Figure 27: RK4 Method Comparison ( $R = 1 \Omega$ ,  $L = 1 H$ ,  $h = T/10$ )

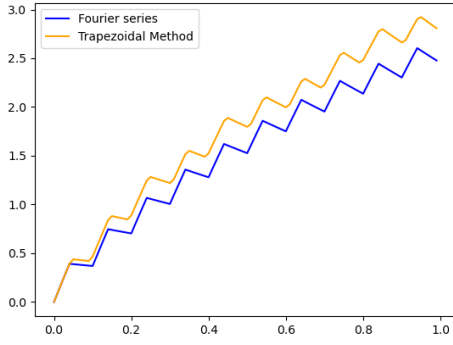


(a) Fourier vs Reverse Euler

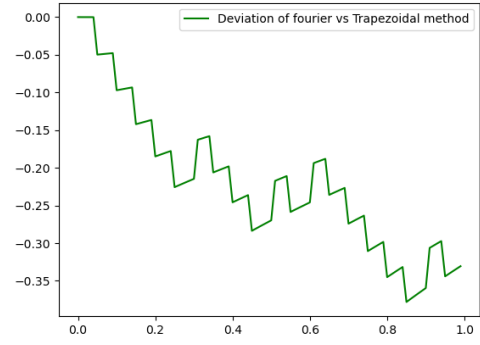


(b) Reverse Euler Error

Figure 28: Reverse Euler Method Comparison ( $R = 1 \Omega$ ,  $L = 1 H$ ,  $h = T/10$ )

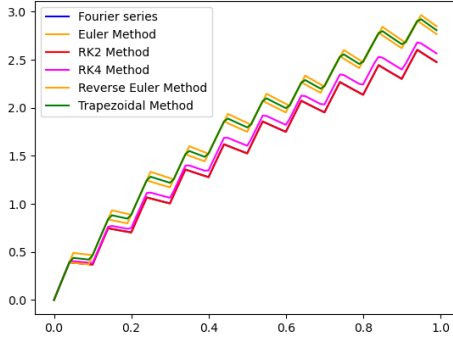


(a) Fourier vs Trapezoidal

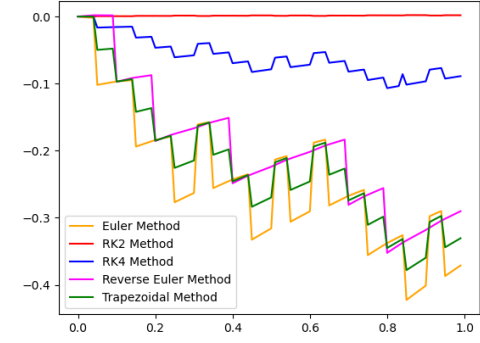


(b) Trapezoidal Error

Figure 29: Trapezoidal Method Comparison ( $R = 1 \Omega$ ,  $L = 1 H$ ,  $h = T/10$ )



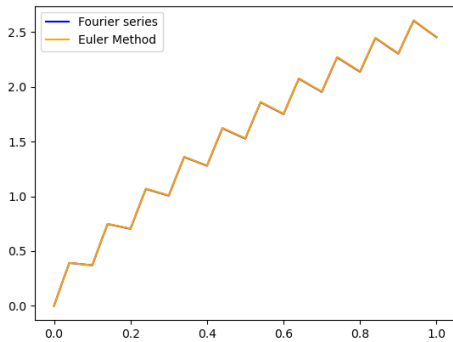
(a) All Methods Comparison



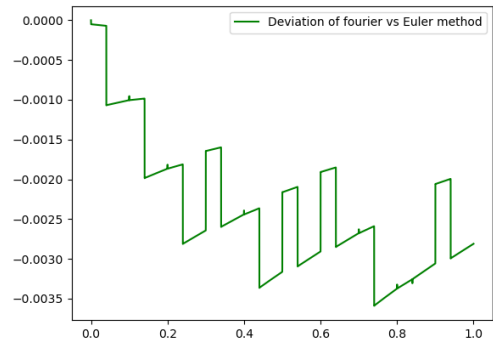
(b) All Methods Error

Figure 30: Comparison of All Methods ( $R = 1 \Omega$ ,  $L = 1 H$ ,  $h = T/10$ )

## 5.2 Time Step $h = T/1000$

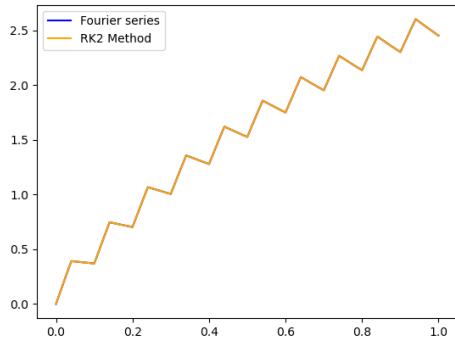


(a) Fourier vs Euler

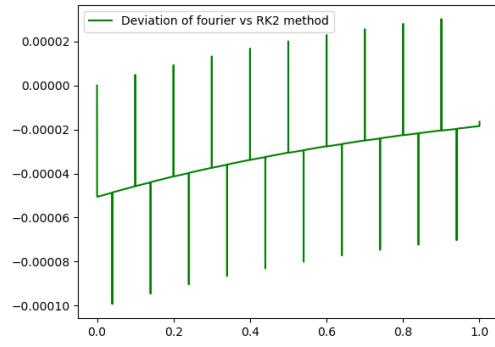


(b) Euler Error

Figure 31: Euler Method Comparison ( $R = 1 \Omega$ ,  $L = 1 H$ ,  $h = T/1000$ )

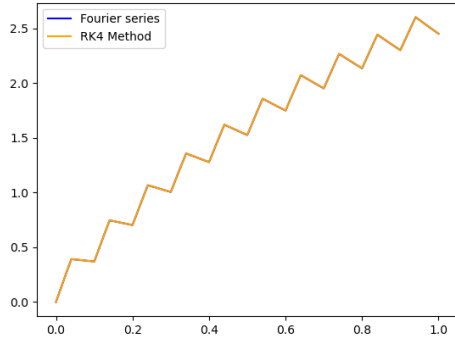


(a) Fourier vs RK2

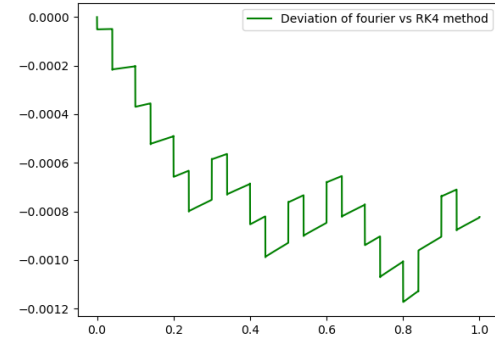


(b) RK2 Error

Figure 32: RK2 Method Comparison ( $R = 1 \Omega$ ,  $L = 1 H$ ,  $h = T/1000$ )

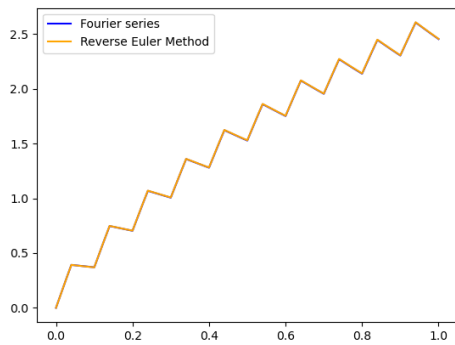


(a) Fourier vs RK4

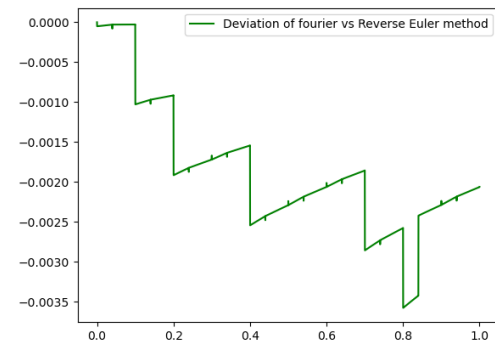


(b) RK4 Error

Figure 33: RK4 Method Comparison ( $R = 1 \Omega$ ,  $L = 1 H$ ,  $h = T/1000$ )

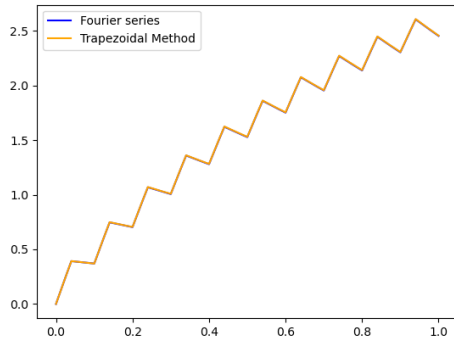


(a) Fourier vs Reverse Euler

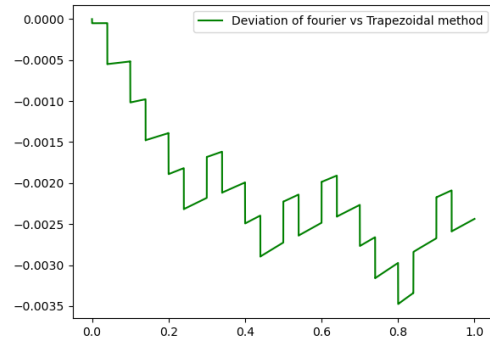


(b) Reverse Euler Error

Figure 34: Reverse Euler Method Comparison ( $R = 1 \Omega$ ,  $L = 1 H$ ,  $h = T/1000$ )

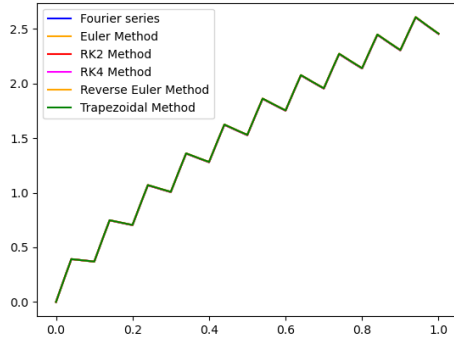


(a) Fourier vs Trapezoidal

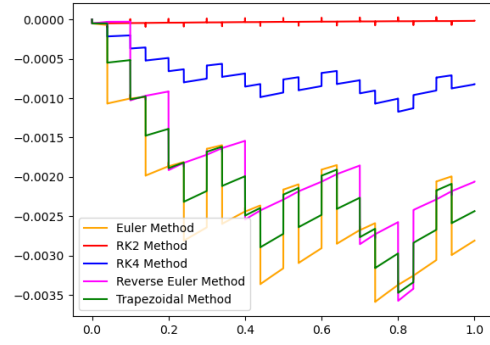


(b) Trapezoidal Error

Figure 35: Trapezoidal Method Comparison ( $R = 1 \Omega$ ,  $L = 1 \text{ H}$ ,  $h = T/1000$ )



(a) All Methods Comparison



(b) All Methods Error

Figure 36: Comparison of All Methods ( $R = 1 \Omega$ ,  $L = 1 \text{ H}$ ,  $h = T/1000$ )

## 6 Results for $R = 0.001 \Omega$

### 6.1 Time Step $h = T/10$

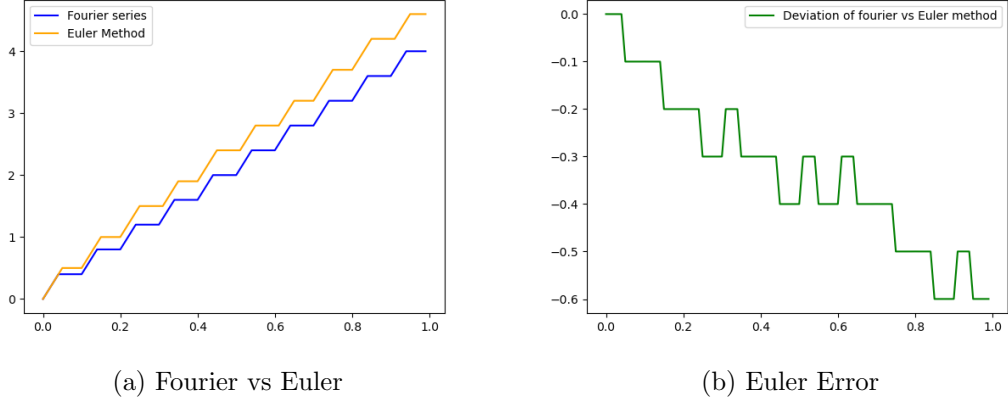


Figure 37: Euler Method Comparison ( $R = 0.001 \Omega$ ,  $L = 1 \text{ H}$ ,  $h = T/10$ )

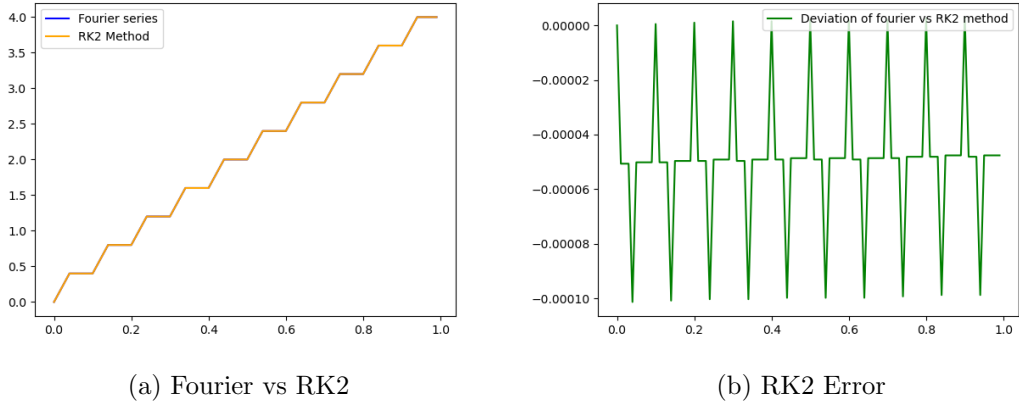
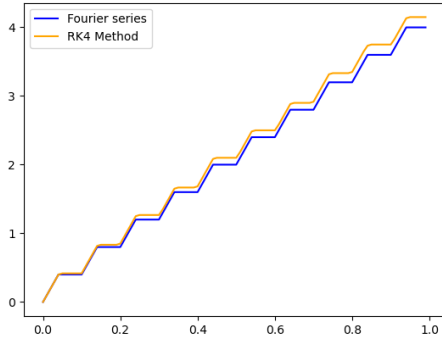
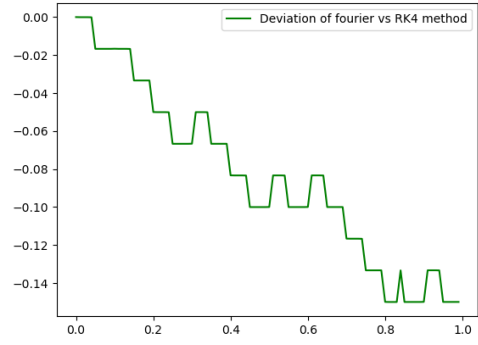


Figure 38: RK2 Method Comparison ( $R = 0.001 \Omega$ ,  $L = 1 \text{ H}$ ,  $h = T/10$ )

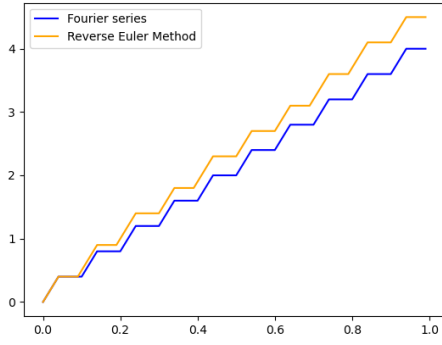


(a) Fourier vs RK4

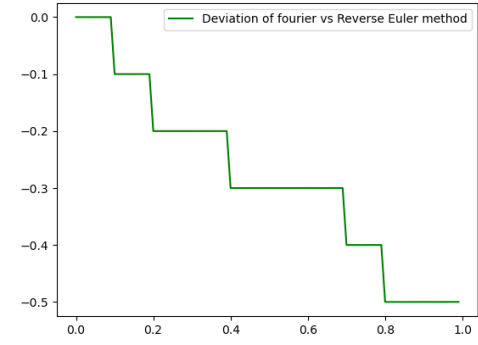


(b) RK4 Error

Figure 39: RK4 Method Comparison ( $R = 0.001 \Omega$ ,  $L = 1 \text{ H}$ ,  $h = T/10$ )

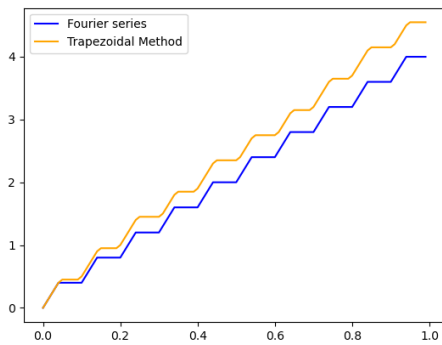


(a) Fourier vs Reverse Euler

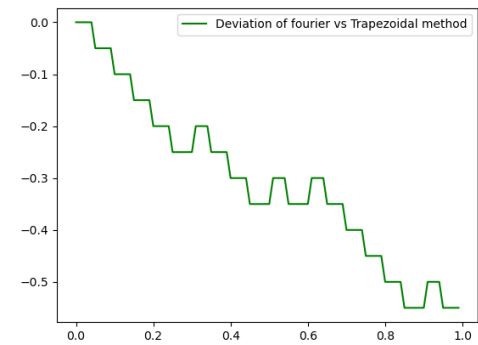


(b) Reverse Euler Error

Figure 40: Reverse Euler Method Comparison ( $R = 0.001 \Omega$ ,  $L = 1 \text{ H}$ ,  $h = T/10$ )

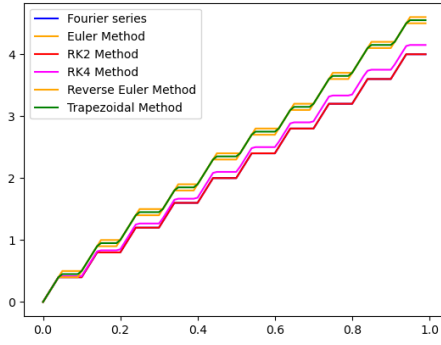


(a) Fourier vs Trapezoidal

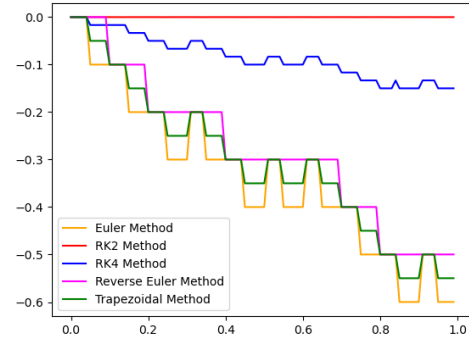


(b) Trapezoidal Error

Figure 41: Trapezoidal Method Comparison ( $R = 0.001 \Omega$ ,  $L = 1 \text{ H}$ ,  $h = T/10$ )



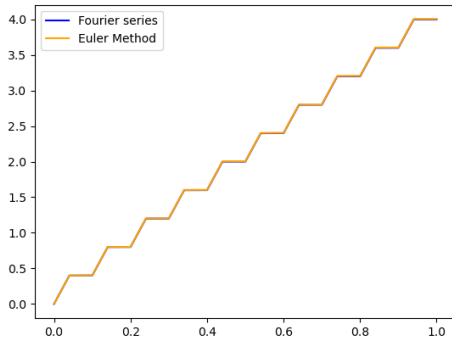
(a) All Methods Comparison



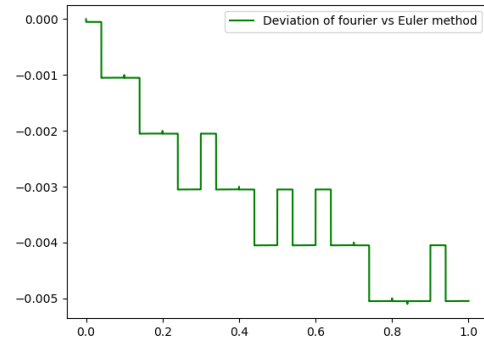
(b) All Methods Error

Figure 42: Comparison of All Methods ( $R = 0.001 \Omega$ ,  $L = 1 \text{ H}$ ,  $h = T/10$ )

## 6.2 Time Step $h = T/1000$

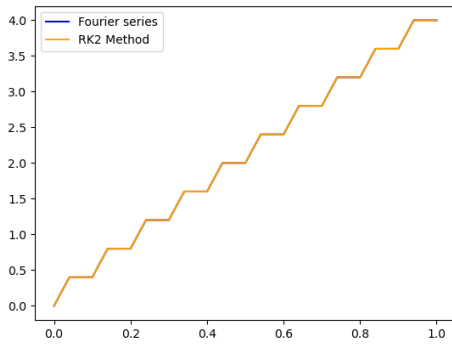


(a) Fourier vs Euler

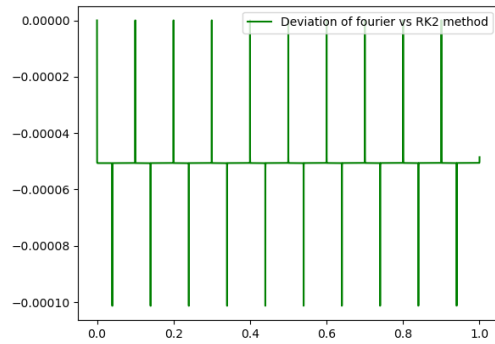


(b) Euler Error

Figure 43: Euler Method Comparison ( $R = 0.001 \Omega$ ,  $L = 1 \text{ H}$ ,  $h = T/1000$ )

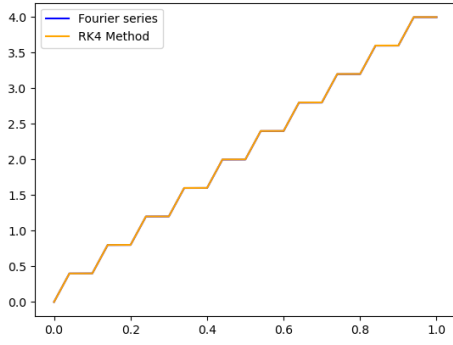


(a) Fourier vs RK2

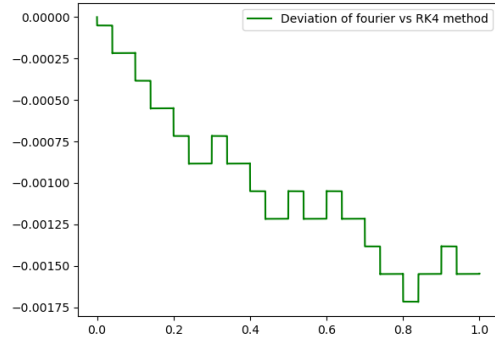


(b) RK2 Error

Figure 44: RK2 Method Comparison ( $R = 0.001 \Omega$ ,  $L = 1 \text{ H}$ ,  $h = T/1000$ )

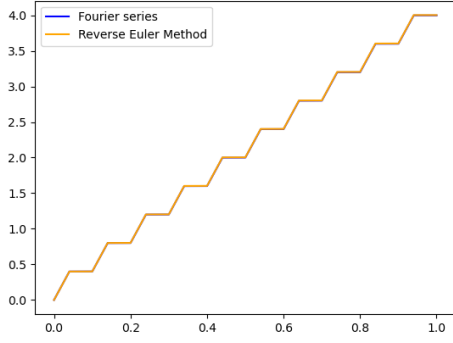


(a) Fourier vs RK4

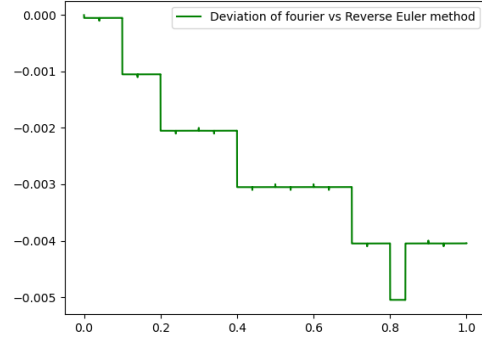


(b) RK4 Error

Figure 45: RK4 Method Comparison ( $R = 0.001 \Omega$ ,  $L = 1 \text{ H}$ ,  $h = T/1000$ )

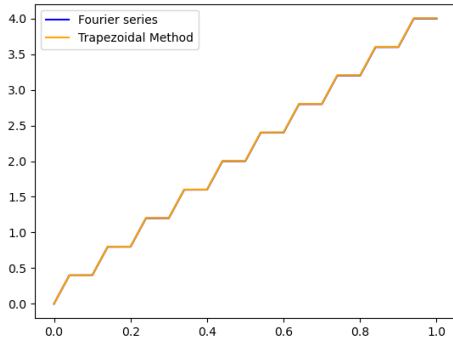


(a) Fourier vs Reverse Euler

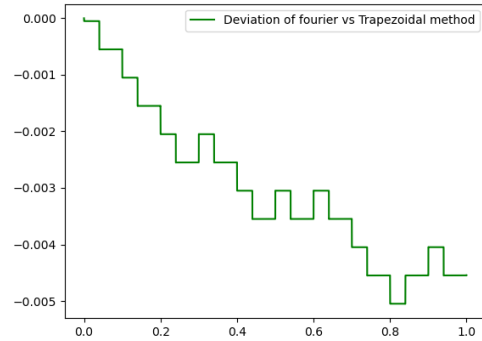


(b) Reverse Euler Error

Figure 46: Reverse Euler Method Comparison ( $R = 0.001 \Omega$ ,  $L = 1 \text{ H}$ ,  $h = T/1000$ )

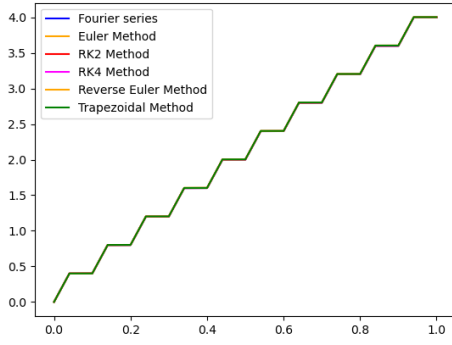


(a) Fourier vs Trapezoidal

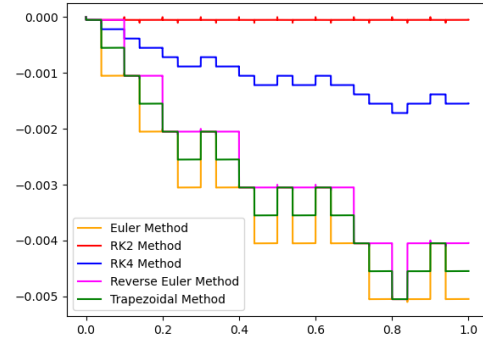


(b) Trapezoidal Error

Figure 47: Trapezoidal Method Comparison ( $R = 0.001 \Omega$ ,  $L = 1 \text{ H}$ ,  $h = T/1000$ )



(a) All Methods Comparison



(b) All Methods Error

Figure 48: Comparison of All Methods ( $R = 0.001 \Omega$ ,  $L = 1 \text{ H}$ ,  $h = T/1000$ )

## 7 Summary

This report has presented a comprehensive visual comparison of various numerical methods with the Fourier series solution for an RL circuit with PWM input. The plots show both the direct comparison of current values and the error between each numerical method and the Fourier series solution.

The comparison was conducted across different resistance values ( $1000 \Omega$ ,  $100 \Omega$ ,  $1 \Omega$ ,  $0.001 \Omega$ ) and time steps ( $T/10$ ,  $T/1000$ ), while keeping the inductance constant at  $1 \text{ H}$ . The duty cycle was set at  $0.4$  and the amplitude at  $10.0$ .